QN : 1 – Riemann Summations

serial

time taken : 1.0719999999999998E-003 INTEGRAL = 2127.3904055871162

parallel

nthreads : 2 time taken : 7.4539999999999997E-003 INTEGRAL = 2127.3904055871176

nthreads : 4 time taken : 1.0883000000000000E-002 INTEGRAL = 2127.3904055871176

nthreads : 8 time taken : 5.1529999999999996E-003 INTEGRAL = 2127.3904055871176

QN : 1 – Monte Carlo Method

serial ==

N = 100 INTEGRAL = 2108.4610198233872 +/- 184.56367000944721 time 4.0000000000000972E-003 ms

N = 1000 INTEGRAL = 2043.6867138945172 +/- 58.889186717553692 time 1.8999999999999920E-002 ms

N = 10000 INTEGRAL = 2037.3386738372242 +/- 18.300821687422797 time 0.13300000000000009 ms

N = 10000 INTEGRAL = 2026.5401046875184 +/- 18.279808521050096 time 0.14599999999999985 ms

N = 100000 INTEGRAL = 2044.2449720014970 +/- 5.8215882645441823 time 1.3390000000000000 ms

N = 1000000 INTEGRAL = 2038.0144834717426 +/- 1.8393861558654478 time 13.481000000000000 ms

N = 10000000 INTEGRAL = 2039.3311171795569 +/- 0.58170664311019360 time 131.35499999999999 ms

parallel ==

nthreads = 2

N = 100 INTEGRAL = 1852.2455711957582 +/- 168.70798094533396 time 0.12399999999999997 ms

N = 1000 INTEGRAL = 2039.0513175805527 +/- 58.162322036680493 time 0.11399999999999995 ms

N = 1000 INTEGRAL = 2004.1104869816136 +/- 58.018050127189866 time 0.11400000000000017 ms

N = 10000 INTEGRAL = 2011.4402101209339 +/- 18.182198770234791 time 0.18400000000000014 ms

N = 100000 INTEGRAL = 2032.0471481824816 +/- 5.8089757201551979 time 1.3119999999999998 ms

nthreads = 4

N = 100 INTEGRAL = 2077.4633109637366 +/- 179.74184944164836 time 0.15600000000000011 ms

N = 1000 INTEGRAL = 1955.2617270074206 +/- 56.808444532418832 time 0.16200000000000003 ms

N = 1000 INTEGRAL = 2101.9213554166181 +/- 60.538199400304464 time 0.15299999999999991 ms

N = 10000 INTEGRAL = 2054.2490325160479 +/- 18.502908147351164 time 0.22300000000000011 ms

N = 100000 INTEGRAL = 2028.0345437203182 +/- 5.7935747014630703 time 0.87999999999999967 ms

nthreads = 8

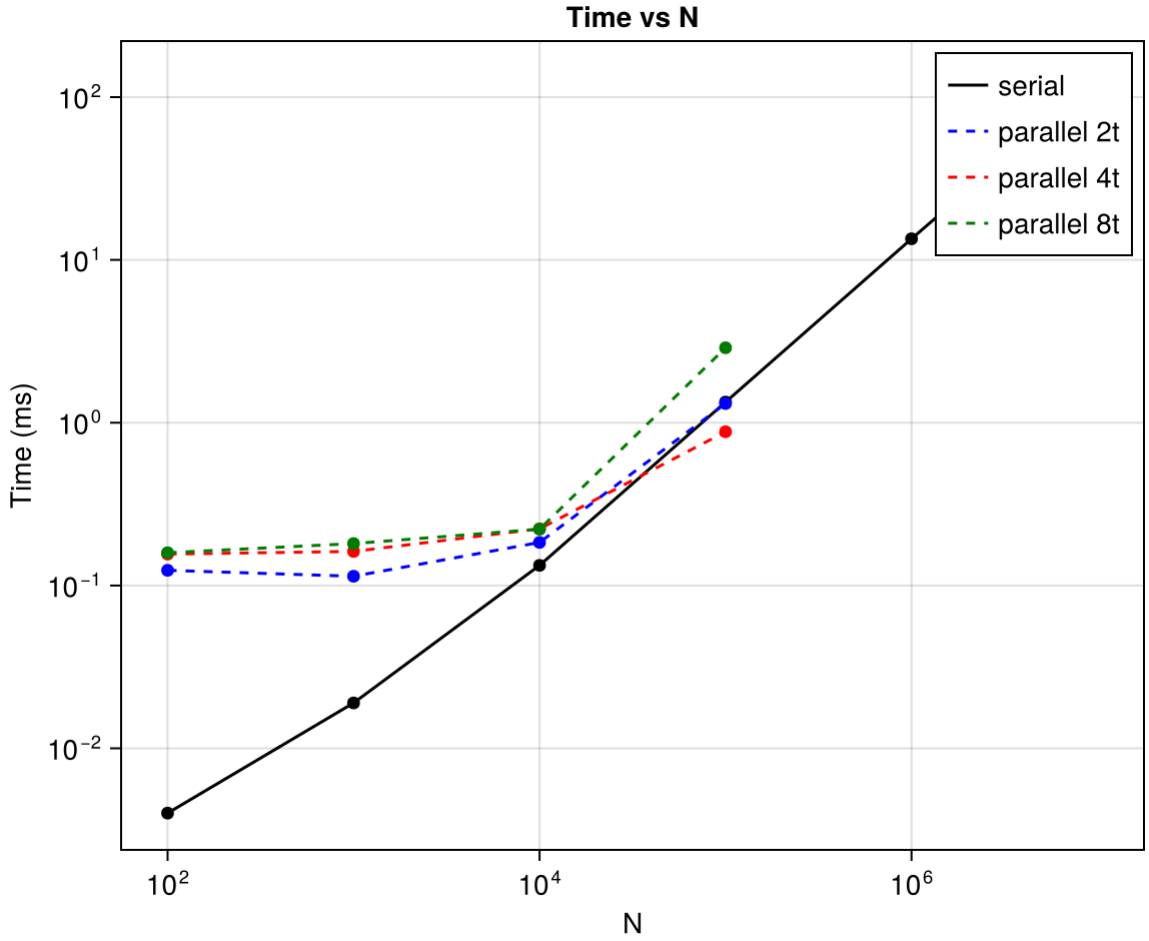
N = 100 INTEGRAL = 2311.9000991021162 +/- 194.27914907891721 time 0.15899999999999986 ms

N = 1000 INTEGRAL = 2030.0240736453845 +/- 57.195661850541541 time 0.18100000000000016 ms

N = 10000 INTEGRAL = 2038.9643897270339 +/- 18.342848505311569 time 0.22199999999999998 ms

N = 10000 INTEGRAL = 2045.8338311042230 +/- 18.439449330627259 time 0.21900000000000000 ms

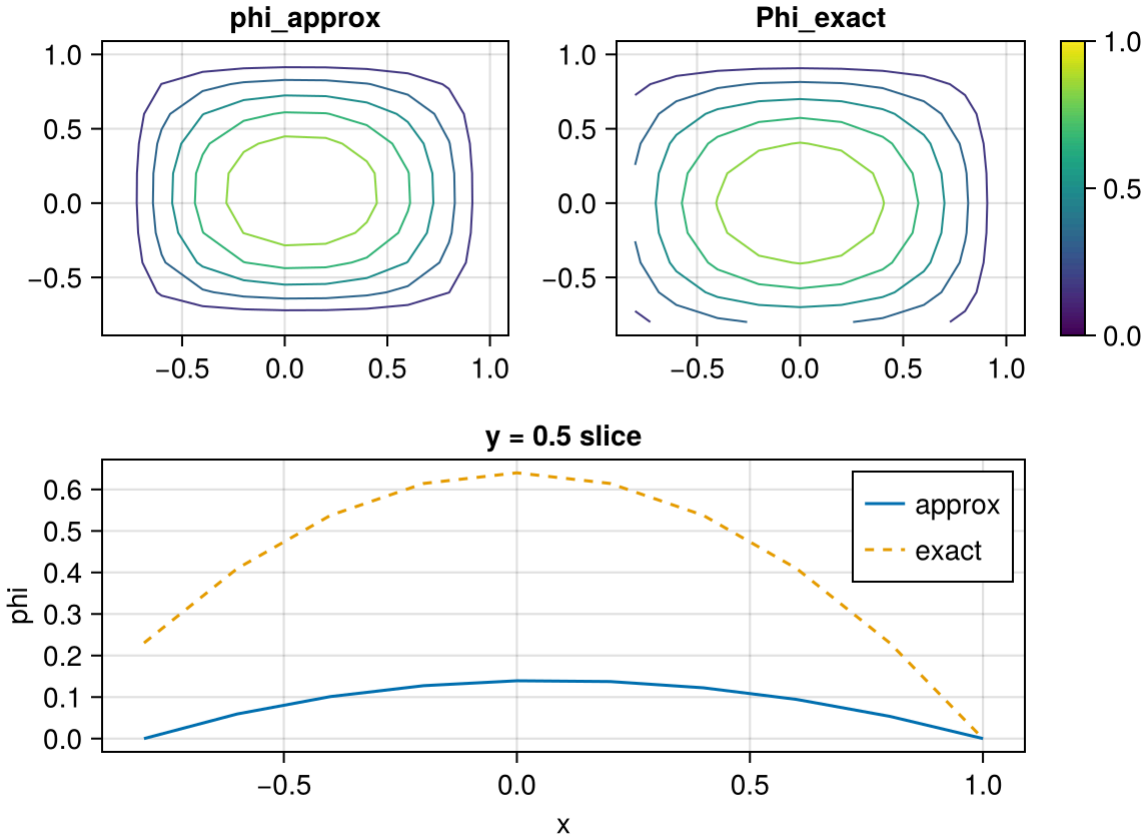
N = 100000 INTEGRAL = 2036.5872651121808 +/- 5.8029821173976046 time 2.8860000000000006 ms



Q NO 2 Poisson Equations

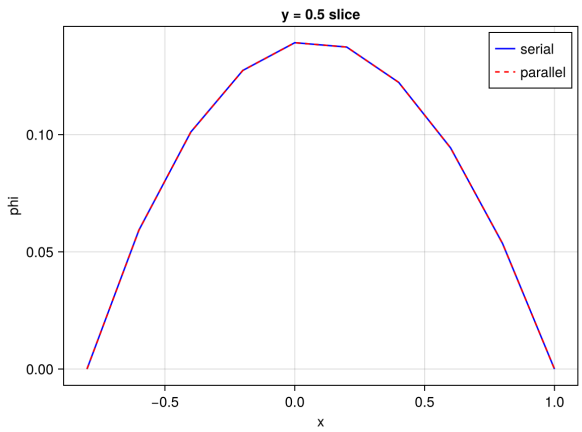
$\Delta = \Delta y = \Delta x = 0.1$

Number of iteration required covergence with error less than 1% = Plot of numerical and analytical solution of phi using serial program



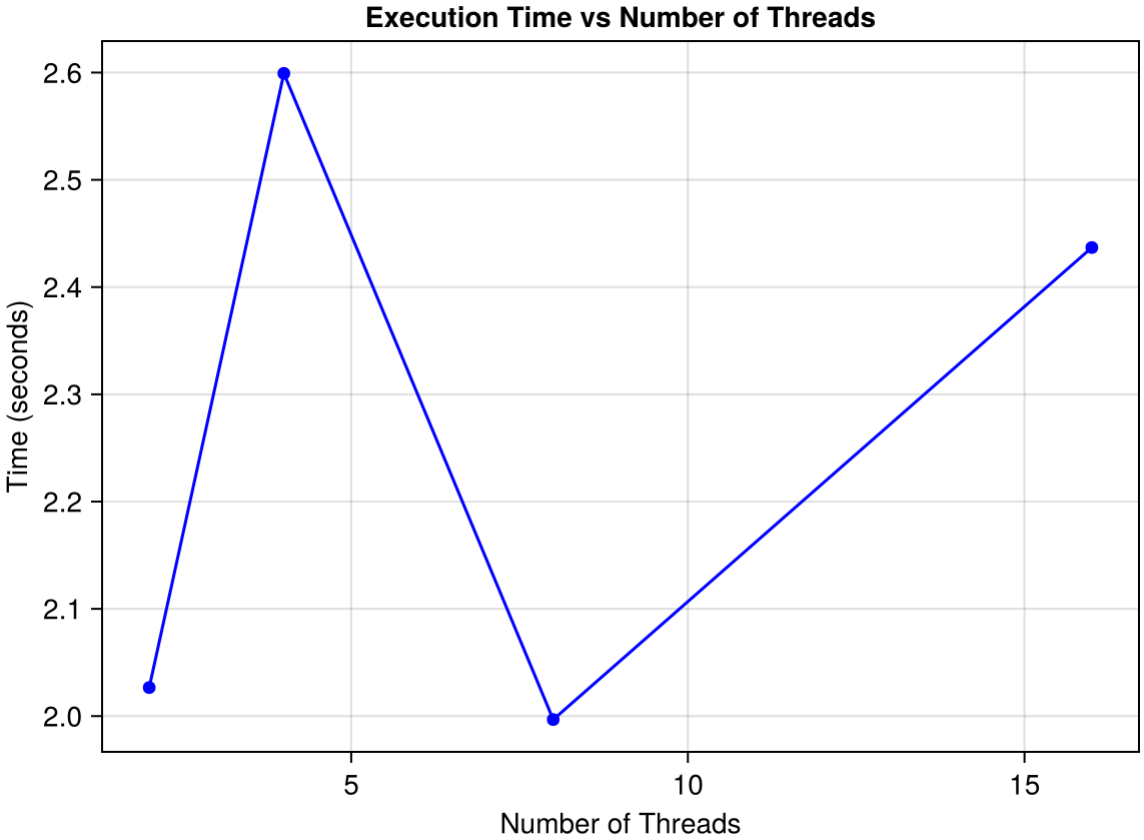
for verification that parallel and serial implementation give same result graph at y = 0.5 slice (Φ vs x) is plotted for both serial and parallel program

at y = 0.5 slice



The OMP_SET_NUM_THREAD is set to 8 using the following environment variable

!sh export OMP_SET_NUM_THREAD=8

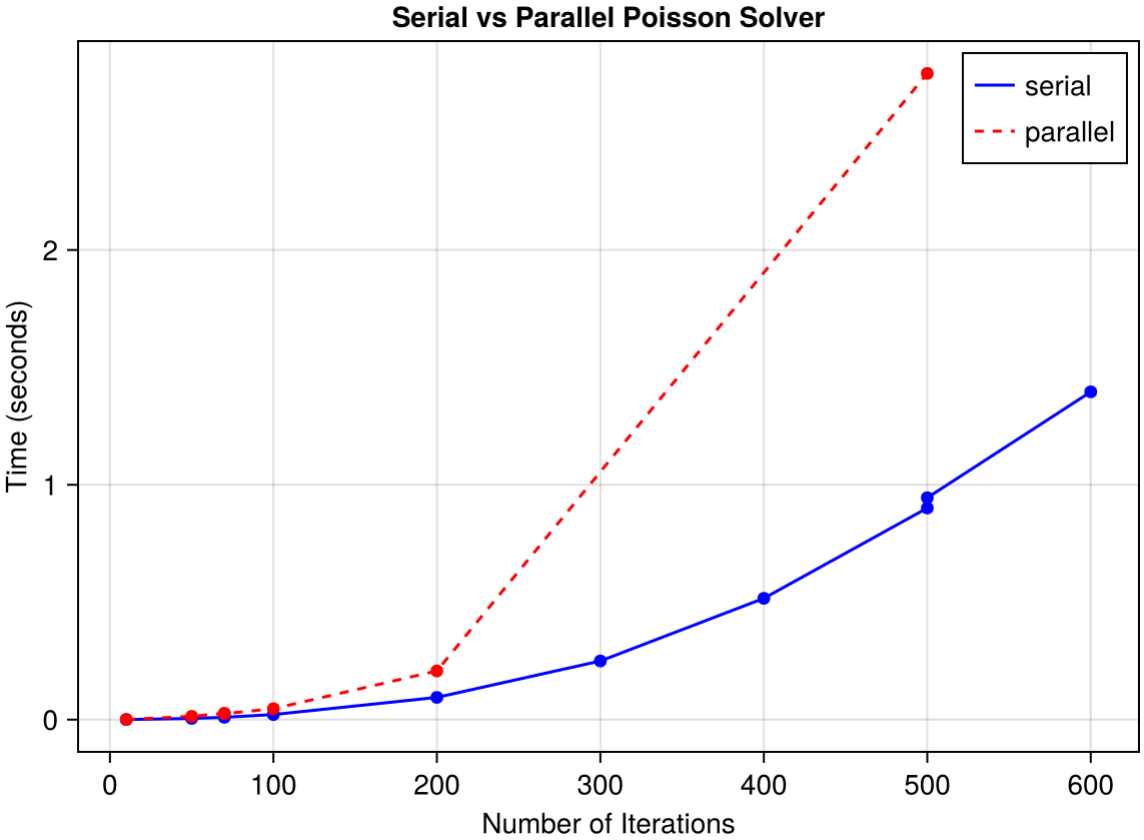


No of threads: 2 Time taken : 2.0265689999999998

No of threads: 4 Time taken : 2.5992540000000002

No of threads: 8 Time taken : 1.9967379999999999

No of threads: 16 Time taken : 2.4368750000000001


Serial vs Parallel Poisson Solver
serial
parallel
Time (seconds)
0
1
2
0
100
200
300
400
500
600
Number of Iterations